

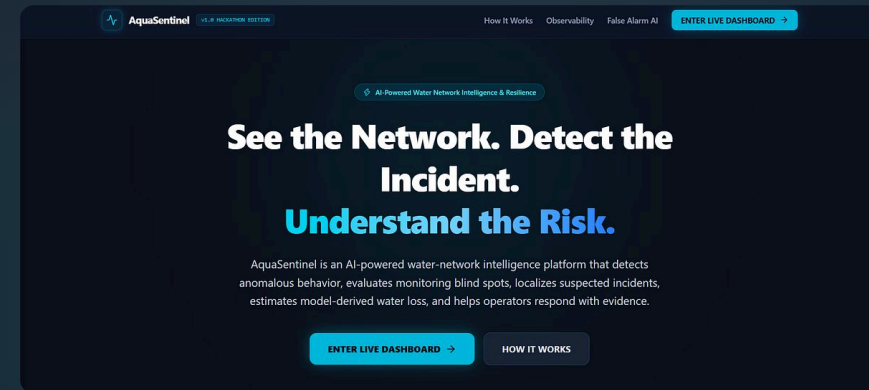
AquaSentinel

AI-Powered Water Network Intelligence & Resilience

"The most important infrastructure in a city is the infrastructure you cannot see."

HACKINDIA SPARK-12 · JAIPUR 2026

TEAM BANKAI





The Problem No One Can See



Invisible Failures

Leaks, bursts & faults happen underground — silently, continuously



Late Detection

Traditional alerting: no spatial context, no network topology awareness



False Alarms

A sensor spike isn't always a leak — but systems treat it that way

The Right Question Isn't "Is Something Wrong?"

AquaSentinel asks five harder questions:

→ Where?

Topology-aware graph localisation

→ How observable?

Blind-spot scoring before you trust the alert

→ Impact?

Model-derived water-loss estimation in LPM / Litres

→ Leak or fault?

LEAK_SUSPECTED · SENSOR_FAULT ·
BURST_EVENT

→ Evidence?

Correlated sensor chain + Gemini grounded explanation



❏ Differentiator: Monitor the Network. Monitor the Monitoring.

The AquaSentinel Control Room

Production-quality operator console — live at aquasentinel-rouge.vercel.app

What operators see

Live Network Map

10 sensors · 10 segments · 3 zones (A/B/C)

Incident Console

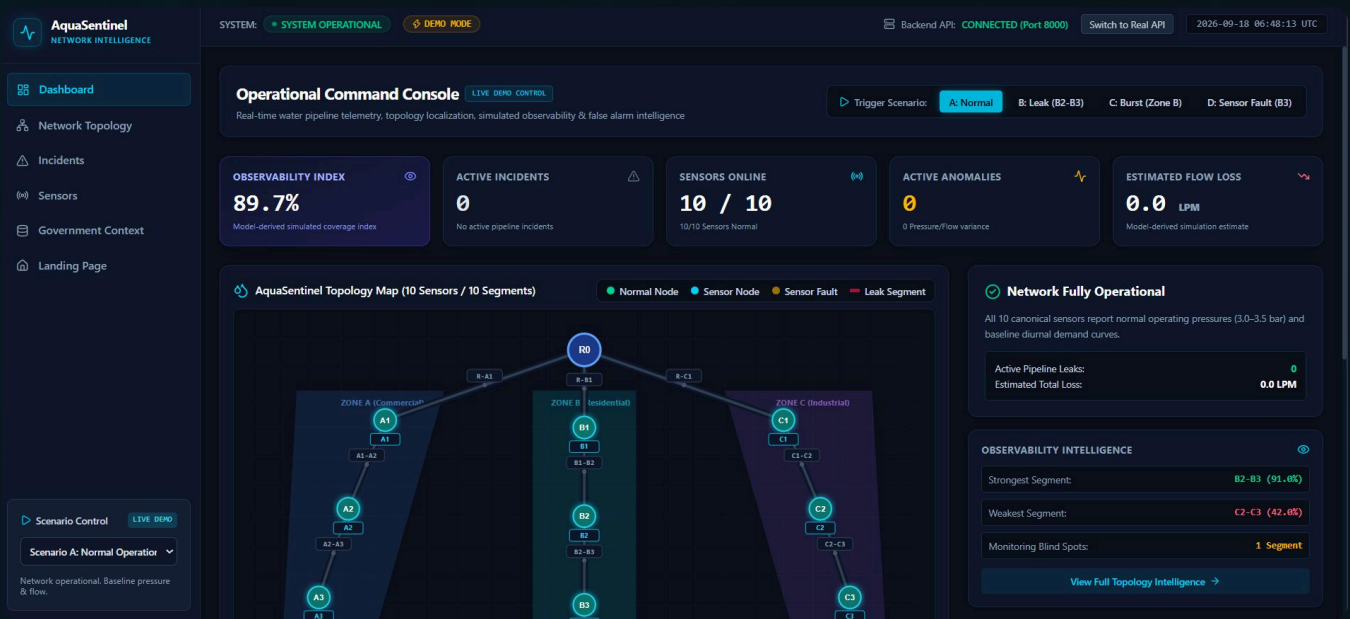
OPEN → ACKNOWLEDGED → RESOLVED lifecycle

Sensor Telemetry

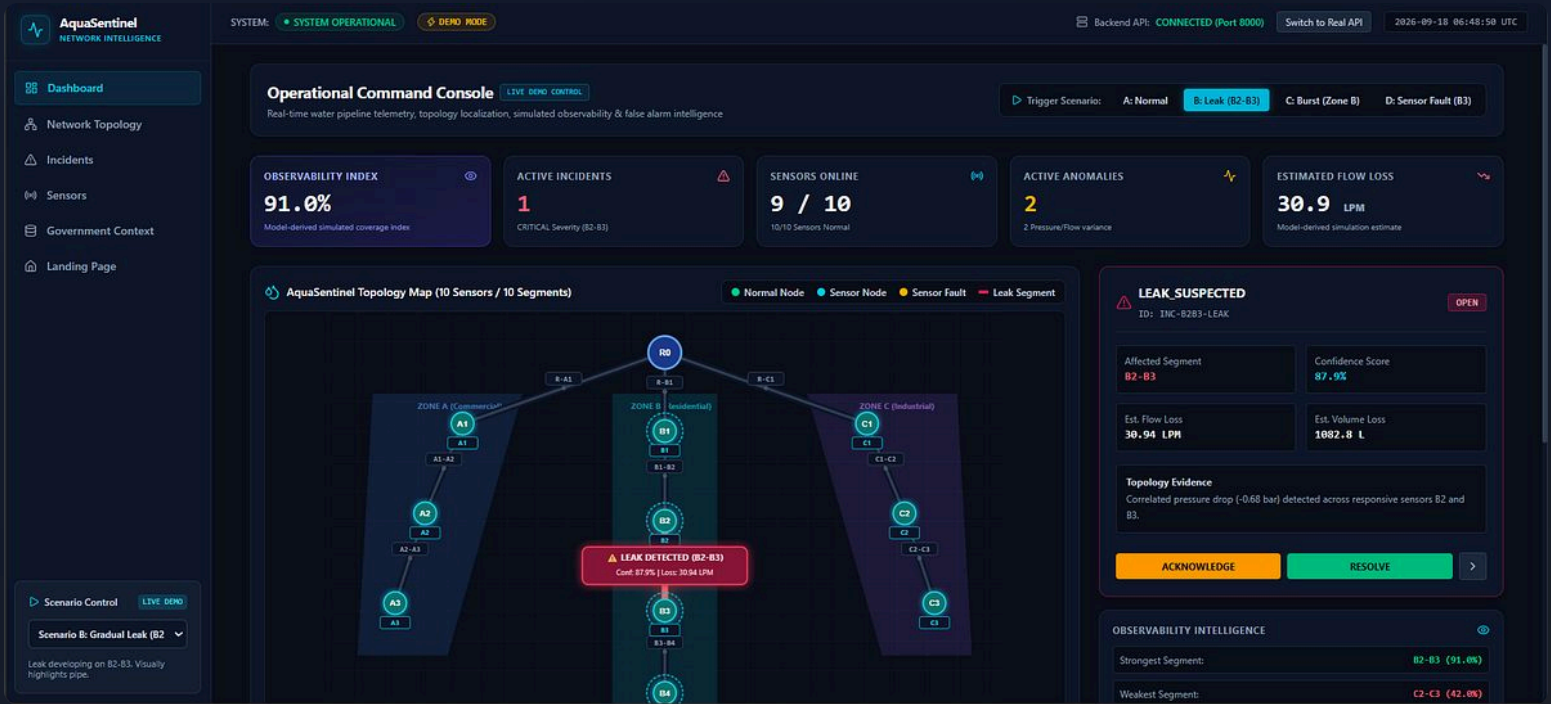
Pressure, flow, statistical scores per node

Observability Score

System-wide blind-spot visibility %



</> NEXTJS 16 · FASTAPI · PYTHON 3.11+



Live Demo: Gradual Leak on B2–B3

89.7%

Observability

System confidence in detection coverage

30.94

LPM Lost

Model-derived flow loss (simulated)

1082

Litres

Estimated cumulative water loss (simulated)

Incident INC-66964E0E

01

Edge pulses red

B2–B3 segment anomaly flagged on network map

02

Correlated evidence

Rolling Z-Score · Rate-of-Change · Isolation Forest

03

LEAK_SUSPECTED

SHA-256 persisted · Gemini 2.5 Flash explanation generated

04

Operator resolves

OPEN → ACKNOWLEDGED → RESOLVED

All figures are model-derived / simulated telemetry

"Not Every Anomaly is a Leak."

False Alarm Demo: B3 Sensor Fault

▲ B3 Pressure

Spikes +2.8 bar · B3 turns amber

✓ B2 & B4 Normal

Neighbours stay cyan/healthy · no propagation

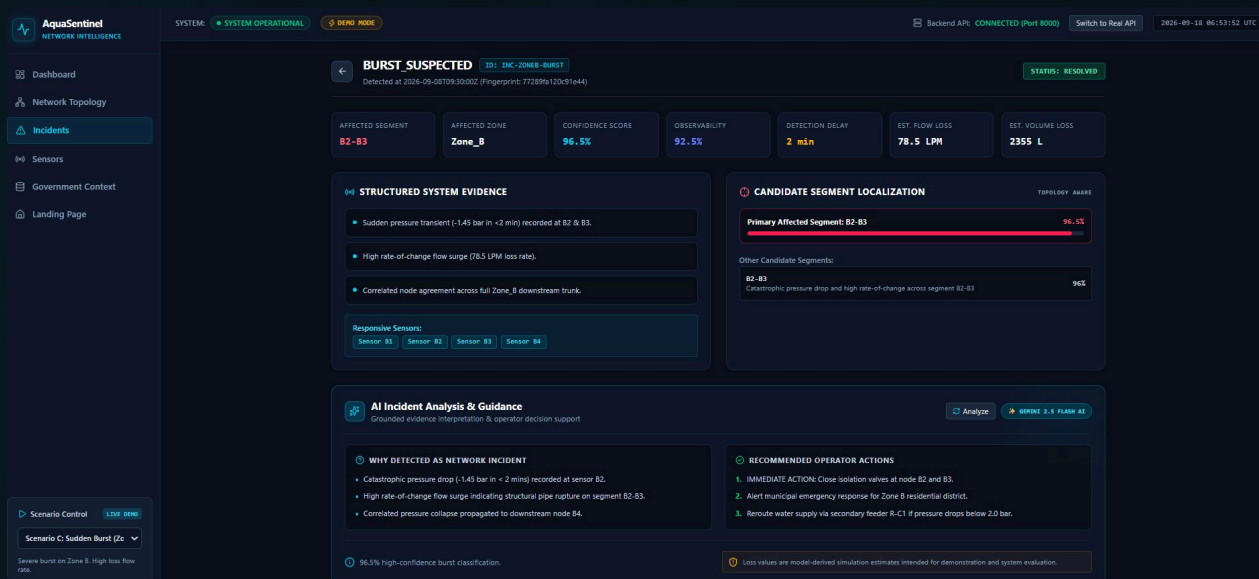
Classification: SENSOR_FAULT

Network topology rules out a
real leak

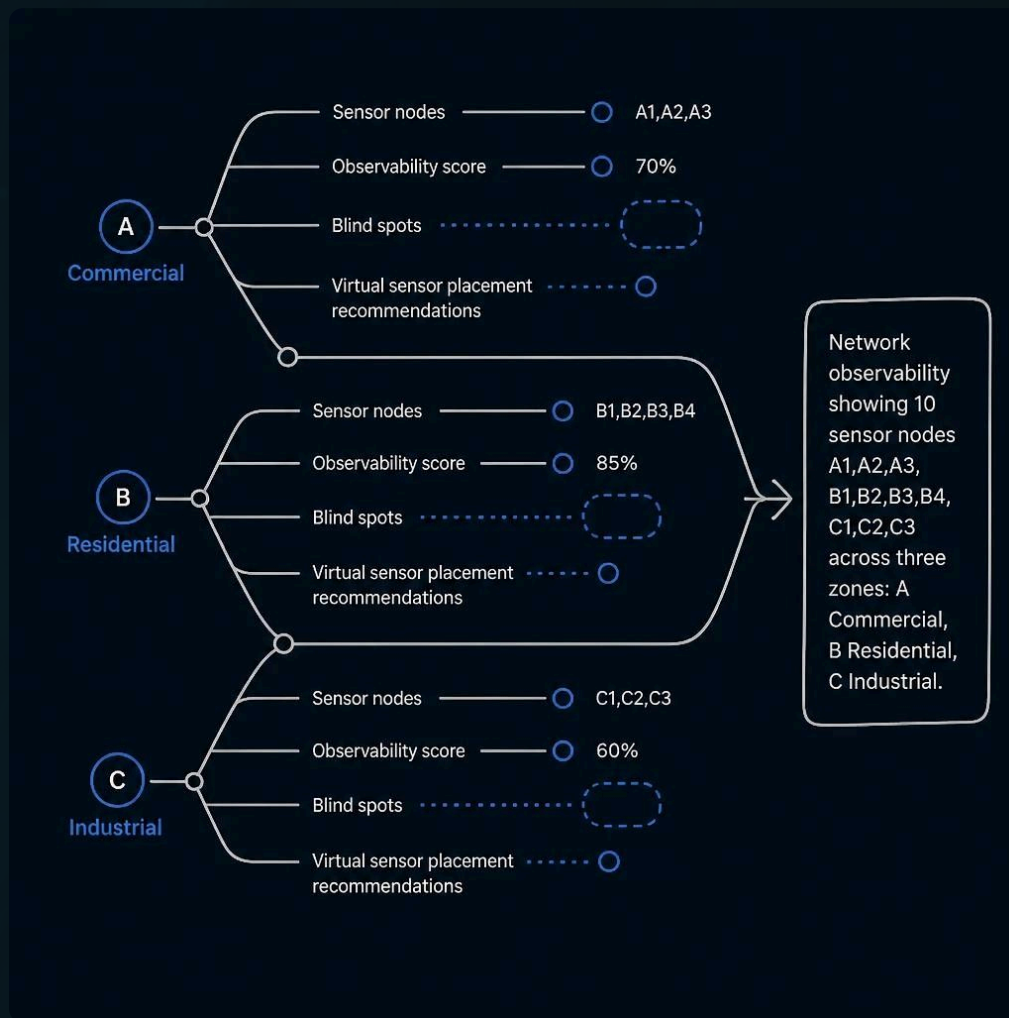
Estimated Water Loss

Zero. No pipeline dispatch
needed.

✓ Spatial context prevents
costly false dispatch



Network Observability — Monitor the Monitoring



Why observability score matters

An alert is only trustworthy if the sensors covering that segment are reliable and dense enough.

1

Segment Scoring

Per-edge observability computed from sensor graph

2

Blind Spot Detection

Under-monitored zones surfaced automatically

3

Virtual Sensor Hints

Roadmap: placement recommendations to close gaps

How It Works



Telemetry

Anomaly
Detection

Observability
Score

Topology
Localisation

Water-Loss &
Classify

Frontend

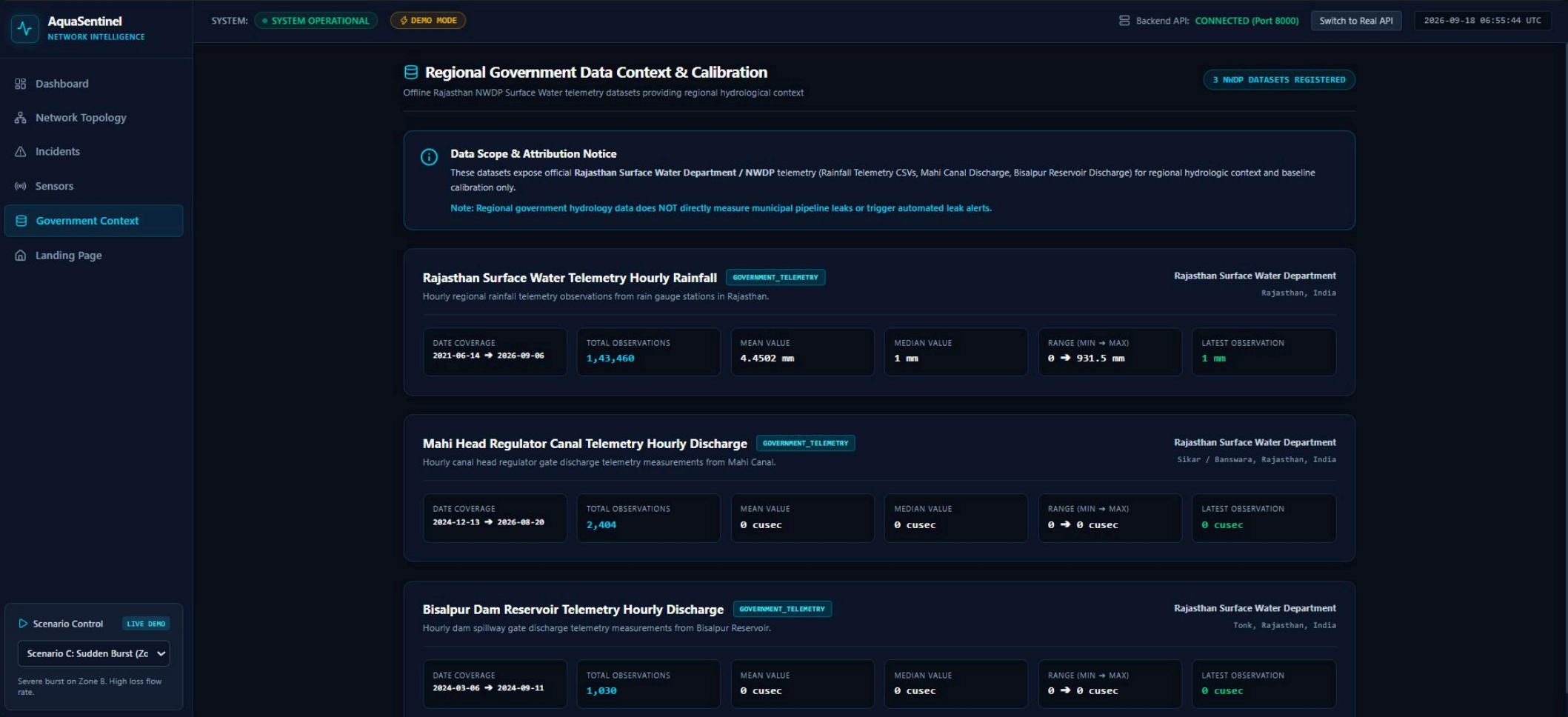
Next.js 16 · React 19 · TypeScript · Tailwind ·
Recharts · SVG

Backend

Python 3.11+ · FastAPI · Pydantic v2 ·
SQLAlchemy 2 · NetworkX · scikit-learn ·
SciPy

Quality

65 pytest tests · lint · type-check · build
validation



AI, Context & What Comes Next

Gemini 2.5 Flash — Explainability Layer Rajasthan Government Data

Offline rainfall telemetry · Mahi Head Regulator Canal discharge · Bisalpur Dam reservoir — calibration & context, not direct leak measurement

Roadmap

-  **Physical Sensors**
ESP32 / Modbus hardware integration
-  **EPANET Model**
Hydraulic simulation & field calibration
-  **WebSocket Streaming**
True real-time telemetry push
-  **GIS Overlays + Alerts**
Shapefiles · SMS / Webhook notifications

Server-side only

Receives structured pre-computed facts · never raw sensor feeds

Structured output

summary · why_detected · recommended_actions · confidence_note · limitations

Deterministic fallback

If Gemini unavailable · system keeps running

See the Network.

Detect the Incident.

Understand the Risk.

Live Prototype

[aquasentinel-
rouge.vercel.app](https://aquasentinel-rouge.vercel.app)

API: [aquasentinel-api-
r00y.onrender.com/docs](https://aquasentinel-api-r00y.onrender.com/docs)

GITHUB: [HACKINDIAXYZ /
BANKAI](#)

Impact in One Line

A city's water network is only as trustworthy as its ability to detect faults, locate them, understand them — and know when not to panic.

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